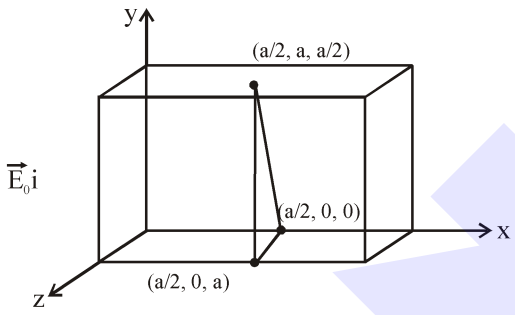
SOLUTION
SECTION-I

Single Correct Answer Type **6 Q. [3 M (–1)]**

1. Ans. (B)

Sol. The flux of y-component of $\vec{E}$ is zero, through this loop

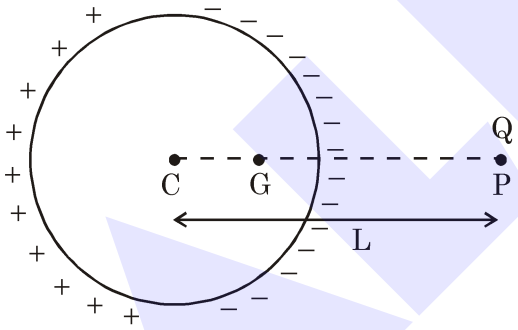


$$\therefore \phi = E_0 \times \frac{1}{2} \times a \times a$$

$$\phi = \frac{E_0 a^2}{2}$$

2. Ans. (A)

Sol.



Inside metallic ball, potential is same everywhere.

$V_G = V_C$

V_C = V due to Q + V due to induced charge

$$= \frac{KQ}{L} + 0$$

$$V_G = V_C = \frac{KQ}{L}$$

3. Ans. (B)

Sol. Use gauss theorem $\phi = \frac{q_{enclosed}}{\epsilon_0}$ = Number of electric field lines

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4. Ans. (B)

Sol. $\rho = \alpha r^2$ [where α is a constant]

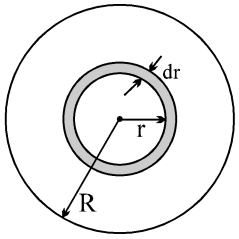
Charge in the shell (element)

$dq = \rho(4\pi r^2 dr) = \alpha(4\pi)r^4 dr$

$\therefore$ Charge enclosed in sphere of radius r, $q = 4\pi\alpha \int_0^r r^4 dr = \frac{4\pi\alpha r^5}{5}$

By Gauss's theorem,

at r = R/2 $\quad (E_{r=R/2})\left[4\pi\left(\frac{R}{2}\right)^2\right] = \frac{4\pi\alpha(R/2)^5}{5\varepsilon_0}$



$E_{r=R/2} = \frac{\alpha R^3}{40\varepsilon_0}$

Total charge enclosed, $Q = \frac{4\pi\alpha R^5}{5}$

$\therefore \quad E_{r=2R} = \frac{4\pi\alpha R^5}{5\varepsilon_0} \times \frac{1}{4\pi(2R)^2} = \frac{\alpha R^3}{20\varepsilon_0}$

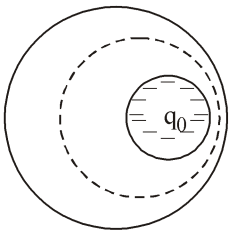
$\frac{E_{r=2R}}{E_{r=R/2}} = 2$

5. Ans. (A)

Sol. Since E_{inside} conductor = 0

$\oint \vec{E}\cdot\vec{ds}$ over guassian surface = 0

$\Rightarrow \oint \vec{E}\cdot\vec{ds} = \frac{q_{in}}{\varepsilon_0} = \frac{q_0 + q_{\text{in on surface}}}{\varepsilon_0} \Rightarrow q_{in}$ on surface $= -q_0$



6. Ans. (C)

Sol. $V_A = V_B$

$r_A \geq r_B$

so, charge density at A > charge density at B

$E_A > E_B$

flux = 0

Multiple Correct Answer Type 12 Q. [4 M (–1)]

7. Ans. (A,B,D)

Sol. Finally potential on all sphere will be equal

$$\frac{KQ_1}{R} = \frac{KQ_2}{2R} = \frac{KQ_3}{3R} \text{ and } Q_1 + Q_2 + Q_3 = (Q) + (-2Q) = -Q \Rightarrow Q_1 = \frac{-Q}{6},\ Q_2 = \frac{-Q}{3},\ Q_3 = \frac{-Q}{2}$$

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8. Ans. (A,C)

Sol.

$\uparrow \frac{\sigma_1}{2\epsilon_0}$

$+ + + + + + + + +$

$\downarrow \frac{\sigma_1}{2\epsilon_0}$

$$E_1 = \frac{8.85\times10^{-9}}{2\times8.85\times10^{-12}} = 500$$

$\uparrow \frac{\sigma_2}{2\epsilon_0}$

$+ + + + + + + + +$

$\downarrow \frac{\sigma_2}{2\epsilon_0}$

$$E_2 = \frac{17.7\times10^{-9}}{2\times8.86\times10^{-12}} = 1000$$

$$E_{net} = \sqrt{\left(\frac{10^3}{2}\right)^2 + \left(10^3\right)^2 + 2\times\frac{10^3}{2}\times10^3\cos 60}$$

$$\sqrt{\frac{7}{4}}\times1000 = 500\sqrt{7}$$

9. Ans. (A, B, C)

Sol. From gauss theorem, $\nabla . \vec{E} = \frac{\rho}{\epsilon_0}$

$$\left(\hat{i}\frac{\partial}{\partial x} + \hat{j}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}\right).\left(x\hat{i} + y\hat{j} + z\hat{k}\right) = \frac{\rho}{\epsilon_0}$$

that gives uniform density.

10. Ans. (A,B,C)

11. Ans. (A,B)

Sol. $\phi = \vec{E}.\vec{A}$

$\phi_{in} = -ve$ and $\phi_{out} = +ve$

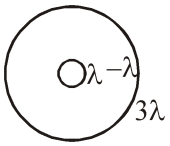
$$Q_{net} = \frac{q_{in}}{\varepsilon_0}$$

$\phi_1 = 0, \phi_2 = E_0a^3, \phi_3 = E_0a^3,\ \phi_4 = 0, \phi_5 = 0, \phi_6 = 0$

$$Q_{net} = \phi_1 + \phi_2 + \phi_3 + \phi_4 + \phi_5 + \phi_6 = 2E_0a^3 \Rightarrow 2E_0a^3 = \frac{q_{in}}{\epsilon_0} \Rightarrow q_{in} = 2\varepsilon_0E_0a^3$$

12. Ans. (A, C, D)

Sol.



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$$\vec{E} = 2\left(\frac{2K\lambda}{r}\right)\hat{r}(r < R)$$

$$\vec{E} = 3\left(\frac{2K\lambda}{r}\right)\hat{r}(r > R)$$

$$\phi_{ABCD} = \phi_{GFEH} + \phi_{BEFD} + \phi_{CGFD} = \frac{q}{4\epsilon_0}$$

13. Ans. (A,D)

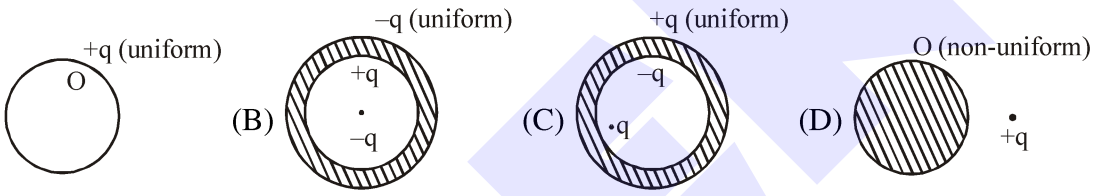
14. Ans. (A,D)

Sol. Due to surface tension bubble try to contract.
Due to surface charge bubble try to expand.

So excess pressure inside the bubble = $\frac{4T}{R} - \frac{\sigma^2}{2\varepsilon_0}$ and electrostatic pressure = $\frac{\sigma^2}{2\varepsilon_0}$

15. Ans. (A,C)

Sol.



16. Ans. (A,C)

17. Ans. (A,D)

Sol. (A) $V_{centre} = \frac{KQ}{a} + \frac{K2\pi\lambda b}{b} = \frac{KQ}{a} + \frac{\lambda}{2\epsilon_0}$

(B) $V_{centre} = 0$ for eathing

$$\frac{KQ'}{a} + \frac{K2\pi\lambda b}{b} = 0$$

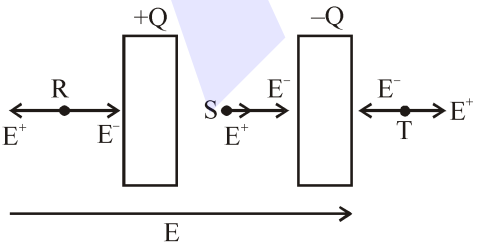
$$Q' = -2\pi\lambda.a$$

(C) $\vec{E}_{inside\ sphere} = \vec{0}$ always (property of conductor)

(D) Conductor without cavity is always equipotential.

18. Ans. (A,D)

Sol.



SECTION-II

Numerical Answer Type Question **1Q.[3M(0)]**

(upto second decimal place)

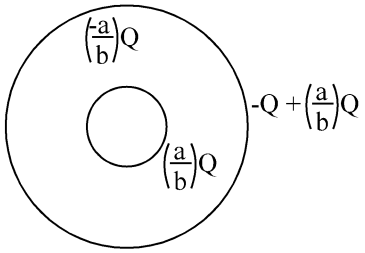
19. Ans. 180

Sol. When outer surface is grounded charge '–Q' resides on the inner surface of sphere 'B'

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Now sphere A is connected to earth potential on its surface becomes zero.
Let the charge on the surface A becomes q

$$\frac{kq}{a}-\frac{kQ}{b}=0\Rightarrow q=\frac{a}{b}Q$$



Consider the figure. In this position energy stored

$$E_1=\frac{1}{8\pi\varepsilon_0 a}\left[\frac{a}{b}Q\right]^2+\frac{Q^2}{8\pi\varepsilon_0 b}+\frac{1}{4\pi\varepsilon_0 b}\left[\frac{a}{b}Q\right](-Q)$$

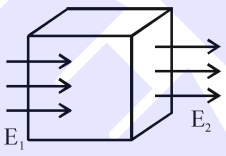
when 'S_3' is closed, total charge will appear on the outer surface of shell 'B'. In this position energy stored $E_2=\frac{1}{8\pi\varepsilon_0 b}\left(\frac{a}{b}-1\right)^2Q^2$ Heat produced $=E_1-E_2=\frac{Q^2a(b-a)}{8\pi\varepsilon_0 b^3}=180$

SECTION-III

Numerical Grid Type (Ranging from 0 to 9) **5 Q. [4 M (0)]**

20. Ans. 3

Sol. Electric field is increasing uniformly according to $|E| = 1 + x$
total flux $= +E_2A - E_1A$

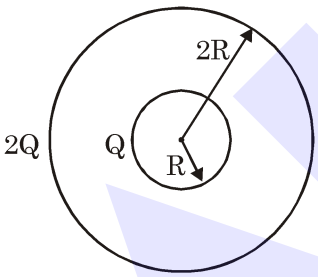


$$\frac{Q_{enc}}{\in_0}=(E_2-E_1)A$$

$= [(1 + x_2) - (1 + x_1)]A$
$= (x_2 - x_1)A = (10\text{ cm})(100\text{ cm}^2) = 10^{-3}\text{m}^3$
$Q_{enc} = 10^{-3} \in_0 = 8.85 \times 10^{-15}$ C

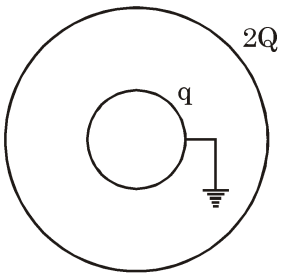
21. Ans. 5

Sol.



$$U_1=\frac{kQ^2}{2R}+\frac{k[2Q]^2}{2\times 2R}+\left[\frac{kQ}{2R}\right]\times 2Q$$

$$=\frac{kQ^2}{R}\left[\frac{1}{2}+1+1\right]=\frac{5}{2}\frac{kQ^2}{R}$$



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$$V_A = 0 = \frac{kq}{R} + \frac{k[2Q]}{2R} = 0$$

$q = -Q$

$$U_2 = \frac{k(Q)^2}{2R} + \frac{k[2Q]^2}{2\times 2R} - \left(\frac{kQ}{2R}\right)\times 2Q$$

$$= \frac{kQ^2}{R}\left[\frac{1}{2} + 1 - 1\right] = \frac{kQ^2}{2R}$$

$$\frac{U_1}{U_2} = 5$$

22. Ans. 4

Sol. For flux to be zero

$q_{in} = 0$

$dq = \rho 4\pi r^2 dr$

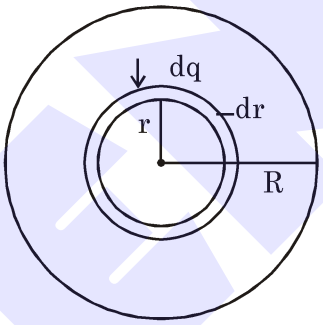
$$dq = \rho_0\left(1 - \frac{nr}{3R}\right)4\pi r^2 dr$$

$$d_{inc} = 4\pi\rho_0 \int_y^R \left(r^2 - \frac{nr^3}{3R}\right)dr$$

$$= 4\pi\rho_0\left(\frac{R^2}{3} - \frac{nR^4}{4\times 3R}\right) = \frac{4\pi\rho_0 R^3}{3}\left[1 - \frac{n}{4}\right]$$

$q_{inc} = 0$ when $1 - \frac{n}{4} = 0$

$n = 4$



23. Ans. 2

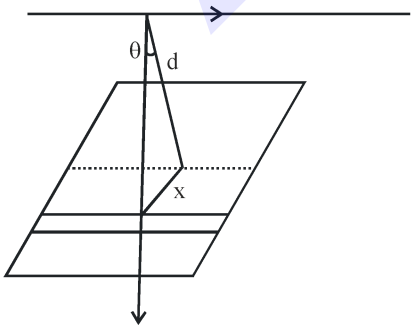
Sol. Force $= \left[\frac{\sigma^2}{2\varepsilon_0}\right]\times A$

$$= \left[\frac{Q}{A}\right]^2 \times \frac{A}{2\varepsilon_0} = \frac{Q^2}{2A\varepsilon_0} = \frac{Q^2}{2a^2\varepsilon_0}$$

$N = 2$

24. Ans. 6

Sol.



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$$d\phi = \frac{\lambda}{2\pi \in_0 \sqrt{d^2+x^2}} \cdot \frac{d}{\sqrt{d^2+x^2}} Ldx$$

$$\phi = \frac{\lambda dL}{2\pi \in_0} \int_{-a/2}^{a/2} \frac{dx}{d^2+x^2}$$

$$= \frac{\lambda(d)(L)}{2\pi \in_0}\left(\frac{1}{d}\right) 2\left[\tan^{-1}\frac{x}{d}\right]_0^{a/2} = \frac{\lambda L}{\pi \in_0}\left(\tan^{-1}\frac{a}{2.\frac{a\sqrt{3}}{2}}\right) = \frac{\lambda L}{6\pi \in_0}$$

SECTION-IV

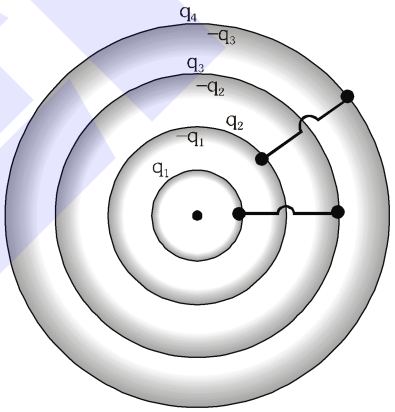
Matrix Match Type (4 × 5) **2 Q. [8 M (for each entry +2(0)]**

25. Ans. (A)-R, (B)-P, (C)-S, (D)-Q

Sol. Let the charge distribution be as shown in the figure below. From Gauss's theorem, we know that facing surfaces of the conductor acquire equal and opposite charges.

$\Rightarrow q_1 + q_3 - q_2 = +4Q$, and .....(i)

Now, $q_2 - q_1 + q_4 - q_3 = -6Q$ .....(ii)



$$V_1 = \frac{1}{4\pi\varepsilon_0}\left[\frac{q_1}{R} + \frac{q_2-q_1}{2R} + \frac{q_3-q_2}{3R} + \frac{q_4-q_3}{4R}\right],$$

$$V_2 = \frac{1}{4\pi\varepsilon_0}\left[\frac{q_1}{2R} + \frac{q_2-q_1}{2R} + \frac{q_3-q_2}{3R} + \frac{q_4-q_3}{4R}\right],$$

$$V_3 = \frac{1}{4\pi\varepsilon_0}\left[\frac{q_1}{3R} + \frac{q_2-q_1}{3R} + \frac{q_3-q_2}{3R} + \frac{q_4-q_3}{4R}\right] \text{and}$$

$$V_4 = \frac{1}{4\pi\varepsilon_0}\left[\frac{q_1}{4R} + \frac{q_2-q_1}{4R} + \frac{q_3-q_2}{4R} + \frac{q_4-q_3}{4R}\right]$$

From $V_1 = V_3$, $q_1 = -\frac{q_2}{3}$; From $V_2 = V_4$, $q_2 = -\frac{q_3}{2}$

On solving eq. (i), $q_1 = \frac{2Q}{5}, q_2 = -\frac{6Q}{5}$ and $q_3 = \frac{12Q}{5}$

Substituting these values in Eq.(ii), $q_4 = -2Q$

For (A) Charge on the inner surface of 3rd conductor = $-q_2 = \frac{6Q}{5}$

For (B) Charge on 4th conductor $= q_4 - q_3 = -2Q - \frac{12Q}{5} = \frac{-22Q}{5}$

For (C) Potential of conductor 1, $V_1 = \frac{-3Q}{40\pi\varepsilon_0 R} = V_3$

For (D) Potential of conductor 2, $V_2 = -\frac{Q}{8\pi\varepsilon_0 R} = V_4$

26. Ans. (A) → (Q,R); (B) → (R,S); (C) → (P,Q); (D) → (P,S)

Sol. Since cavity is spherical charge induces uniformly on its inner surface but an outer surface charge distribution is non-uniform.

Due to point charge electric field is radially outwards.

Near conducting surface net electric field is always perpendicular to the conductor.

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